

Who Let the Dogs Out? Auditing Dogwhistle Coverage in Hate Speech Benchmarks Across Target Groups

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1 Introduction

Social media platforms harbor a growing number of individuals and organizations who use these networks to disseminate hateful, extremist messaging. This content typically violates the terms of service for these sites. Various strategies are used to reduce the proliferation of hate speech, one of which is moderation: filtering, reviewing, and managing controversial content. The use of such systems can greatly curb overt hate speech; however, it can also drive users who wish to communicate bigoted sentiment to use increasingly obfuscated language, known as dogwhistles. [11] [4] [12] [1]. This lexicon of covert expressions, phrases, and emojis that act as shorthand for hateful messages is characterized by the ability to simultaneously convey one innocuous meaning to a broad audience, and another hateful message to a smaller receptive in-group. Within this category of language, classifications can be derived along the axis of transparency to an uninformed audience.

These expressions abound in political campaigns, mass media, everyday language, and, of particular interest to this study, social media platforms. Social networks such as Twitter or YouTube employ automated moderation systems to identify and remove hate speech. However, the task of identifying toxic language is non-trivial: prior work shows that machine-learning based classifiers often fail on coded language [9], [7]. These studies identified training dataset composition as a driving factor behind whether a model properly classifies posts containing dogwhistles. Additionally, they also observed performance variance across different target groups.

We build upon this body of work to examine the coverage gaps of two benchmark datasets (HateXplain [8] and Measuring Hate Speech [6]). This work is significant because benchmarks that systematically underrepresent the coded language associated with certain communities will produce classifiers that provide those communities less protection — a disparate impact violation that originates in dataset construction, not just model training.

1.1 Research Questions

- RQ1:** Do existing hate speech benchmarks have lower coverage of dogwhistle terms as coding sophistication increases?
- RQ2:** When dogwhistle terms do appear in benchmarks, are they correctly labeled as hateful, and does the correct labeling rate vary across groups by coding level?
- RQ3:** Do benchmarks satisfy equalized odds across target groups? That is, is the probability of a dogwhistle term being correctly labeled hateful equal across groups, conditional on coding level?

1.2 Expected Contributions

- EC1:** A conceptual mapping of the Mendelsohn dogwhistle glossary onto a four-level coding sophistication taxonomy which we will publicly release to support future benchmark audits
- EC2:** A quantitative audit of benchmark coverage and annotation quality disaggregated by coding level and target group
- EC3:** Evidence distinguishing collection failure from annotation failure as sources of benchmark gap, which has concrete implications for how future datasets should be constructed

2 Literature Review Outline

There is a sizable body of evidence demonstrating that automated moderation tools struggle with dogwhistles [4] [3] [1] [11]. There are also a number of studies that go one step further to implement new tools to better identify and classify covert hate speech [13] [10]. These studies all point towards the coverage or lack thereof in training data as a key contributor to model performance. However, as of yet, there has been no work performing audits of widely used benchmark datasets in combination with examining downstream classifier performance across target subgroups. This is the gap we seek to fill.

Our literature review covers four areas. First, we survey the landscape of coded and implicit hate speech, including the development and use of dogwhistles in online spaces [2, 16] and the Mendelsohn et al. dogwhistle lexicon that grounds our audit [10]. Second, we review automated moderation systems, covering both their deployment on major social platforms and their documented failures on coded language [4, 5, 10]. Third, we examine hate speech benchmarks, discussing the limitations of existing large-scale datasets [11, 15], our rationale for selecting HateXplain and Measuring Hate Speech as our primary corpus [9, 12], and the Implicit Hate Speech corpus as a secondary comparison [4]. Finally, we ground our fairness analysis in the broader debate over competing fairness definitions, motivating our choice of equalized odds as the appropriate criterion for this audit [1, 6].

For this work, we still need to read papers on the benefits/challenges of automated moderation systems and on which datasets are most commonly used in hate-speech detection.

3 Data

3.1 Dogwhistle Glossary

Our audit operationalizes dogwhistle coverage through the glossary introduced by Mendelsohn et al., which comprises 340 entries and over 1,000 surface forms annotated with target group and dogwhistle type [9]. This is the largest and most systematically annotated dogwhistle lexicon currently available, and provides the ground

truth against which we measure benchmark coverage. We note, however, that the glossary’s own coverage is uneven across target groups: for example, transphobic and white supremacist entries are more densely represented than anti-Asian or anti-Latino ones. To account for this, we normalize all group-level presence rates by the number of glossary entries available for that group, ensuring comparisons reflect benchmark gaps rather than glossary sparsity.

A further limitation is temporal. The Mendelsohn glossary is a snapshot of dogwhistle terminology as documented through 2023, and coded language evolves continuously as communities develop new terms in response to moderation [10]. Our audit therefore cannot claim to measure coverage of the full universe of dogwhistles in use. Crucially, however, this limitation operates in a consistent direction: dogwhistle terms coined after 2023 will be even less represented in benchmarks than those we audit for, meaning our coverage gap estimates are conservative lower bounds rather than overestimates of the problem.

3.2 Benchmarking Datasets for Hate Speech

We construct our benchmark union from two large, human-annotated datasets that include target group labels: HateXplain [8] and Measuring Hate Speech [6].

HateXplain consists of 20,148 posts drawn from Twitter and Gab, each annotated with a three-class hate label (hate, offensive, or normal), one or more target group labels (race, religion, gender, sexual orientation, or miscellaneous), and rationale spans identifying which tokens drove annotator decisions [8]. Measuring Hate Speech contributes 39,565 posts from Twitter, Reddit, and YouTube, annotated with a continuous hate speech severity score and fine-grained target identity labels spanning race/ethnicity, religion, national origin, gender, sexual orientation, age, disability, and political ideology [6]. Together, these datasets provide platform diversity and complementary annotation schemes, one categorical and one continuous.

We note that Davidson et al. [2] remains one of the most widely used datasets for classifier training and evaluation, and its exclusion warrants explanation. The dataset predates much of the dogwhistle terminology we audit for, and critically, it carries no target group annotations, making group-level disaggregation impossible. Its ubiquity as a training resource is itself a finding consistent with our argument: classifiers trained on unannotated benchmarks cannot be evaluated for group-level fairness, let alone improved. We conduct our primary audit on a union of HateXplain and Measuring Hate Speech, reflecting the datasets practitioners most commonly use for classifier training. We then replicate our audit with the Implicit Hate dataset [3] added to the union as a secondary analysis.

4 Methods

4.1 Data Preprocessing

We first union HateXplain and Measuring Hate Speech into a single corpus, standardizing annotation format across datasets. Because Measuring Hate Speech uses a continuous severity score while HateXplain uses discrete labels, we follow Kennedy et al.’s recommended threshold of 0.5 as our primary binarization cutoff, then conduct a sensitivity analysis across neighboring values to verify findings are not threshold-dependent. For posts that have multiple

target labels, we will use the label associated with the dogwhistle term. We then deduplicate the union by post ID where available, falling back to exact text matching where IDs have been stripped for privacy. To match dogwhistle terminology against the corpus, we perform case-insensitive string matching against all Mendelsohn surface forms, supplemented by manually verified regex rules for morphological variants of the small subset of terms that plausibly appear in inflected forms in the wild. We acknowledge that surface form matching will produce false positives. Instances where a dogwhistle term appears in a non-hateful context, such as a tweet critiquing rather than deploying a slogan (e.g. "critiques of the All Lives Matter movement). To estimate the false positive rate, we will manually inspect a random sample of 50–100 matched tweets stratified by coding level and report per-level false positive rates as a limitation on our coverage estimates.

4.2 Proposed Taxonomy of Implicit Hate Speech

We organize the Mendelsohn glossary entries into a four-level coding sophistication taxonomy based on the degree of decoding required to connect a term to its target group, then map each Mendelsohn entry to a level based on its annotated dogwhistle type, as summarized in Table 1. We exclude entries labeled as humor or mockery from our primary analysis, as these cases are not subject to clean level assignment. Some require only stereotype knowledge (Level 2) while others require in-group community knowledge (Level 4), and distinguishing between them would require a separate annotation pass. We acknowledge that this exclusion may cause us to undercount coded language targeting Black communities specifically, since mock-AAVE entries seem to disproportionately carry the humor label in the Mendelsohn glossary; we note further examination is necessary and that this is a directional limitation on our group-level coverage estimates.

4.3 Audits and Metrics

4.3.1 Coverage Audit (RQ1). For each surface form in the Mendelsohn glossary, we compute three metrics aggregated by coding level and target group. *Presence rate* is a binary indicator of whether a surface form appears anywhere in the benchmark union, aggregated to a proportion across all forms at a given level or within a given target group. *Type coverage* measures what proportion of Mendelsohn dogwhistle categories (rather than individual surface forms) appear at least once, capturing whether entire categories of coded language are absent from benchmarks rather than just specific terms. *Token frequency* records how many times each matched surface form appears across the union; a term appearing once versus hundreds of times has meaningfully different implications for whether a classifier trained on that benchmark could learn to detect it.

4.3.2 Annotation Quality Audit (RQ2). For surface forms that do appear in the benchmark union, we examine their assigned labels. We compute the *correct labeling rate* (the proportion of dogwhistle-containing posts labeled hateful) disaggregated by coding level and target group. This is our core annotation failure metric and requires no external resources. We further decompose matched posts into three cases: Case A (present and labeled hateful — correct), Case B (present but labeled non-hateful — annotator failure), and Case C

Table 1: Proposed taxonomy of implicit hate speech coding sophistication, mapped to Mendelsohn et al. dogwhistle types [9]. Color intensity reflects explicitness: from direct slurs (L1) to maximally opaque coded language (L4).

Level	Description	Mendelsohn Type(s)	Examples
Level 1 Explicit slurs	Target group named directly; no decoding required. Baseline.	Direct slur / explicit label	[direct slurs]
Level 2 Stereotype-based	Requires cultural stereotype knowledge to connect term to group.	Stereotype-based target group label; stereotype-based descriptor; phonetic-based label	“ <i>anchor babies</i> ” (anti-Latino), “ <i>blueish</i> ” (antisemitic)
Level 3 Concept / policy	Requires ideological or political context; group never named, even implicitly.	Concept (policy); concept (values); concept (other); representative (Bogeyman)	“ <i>All Lives Matter</i> ,” “ <i>abolish birthright citizenship</i> ,”
Level 4 Persona signals	No recoverable semantic path to target group; requires in-group membership to decode.	Persona signal (symbol, shared culture, self-referential); arbitrary target group	“ <i>alt-right</i> ,” “ <i>biological woman</i> ,”

(absent from the union entirely — collection failure). We report the Case B / (Case A + Case B) ratio as a direct measure of annotator failure, independent of collection gaps. The relative proportion of Case B to Case C across levels and groups distinguishes whether the benchmark gap is primarily a collection problem or an annotation problem, with distinct implications for how future datasets should be constructed.

4.3.3 Disparity Analysis (RQ3). To assess disparate impact under an equalized odds framework [5], we test whether coverage and annotation quality metrics are distributed evenly across target groups conditional on coding level. Specifically, we compute *coverage gap between groups at the same level*. For example, we compute whether the Level 3 presence rate for antisemitic terms differs significantly from the Level 3 presence rate for anti-Latino terms. We also compute *labeling rate disparity across groups*, applying the same comparison to annotation quality. Groups for whom both metrics are lower at higher coding levels bear a compounded disadvantage: their coded language is both less likely to be collected and less likely to be correctly labeled when it is. We treat this interaction between coding level and target group as our core fairness finding.

5 Project Timeline

See table 2.

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Table 2: Project milestones and responsibilities by week.

Wk	Reva	Ryan
6	Finalize humor/mockery exclusion rules; document L1–L4 lookup table from Mendelsohn types	Define primary binarization threshold (0.5); design Case A/B/C classification logic
7	Build preprocessing pipeline (union, dedup, binarize MHS scores, string matching)	Build audit pipeline (presence rate, type coverage, token frequency)
8	Pilot pipeline on sample; manually inspect 50–100 matched posts to estimate per-level false positive rate	Finalize regex rules for morphological variants; validate Case A/B/C classification on pilot sample
9	Run coverage audit (RQ1) on primary union	Run annotation quality audit (RQ2); compute Case B/(A+B) ratios by level and group
10	Sensitivity analysis across binarization thresholds	Replicate audit with ElSherief added to union
11	Disaggregate RQ1/RQ2 results by target group; compute coverage gaps	Disparity analysis (RQ3); statistical significance testing
12	Present, draft final report	Present, draft final report
13	Joint: revision, figures, final report submission	

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